Syed Reebal Faakhir Andrabi

Third Year Undergraduate, Department of Civil Engineering

SRFA07 | Portfolio | reebalf24 | Reebal Andrabi

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2024 - Present	B.Tech	Indian Institute of Technology Kanpur	**8.7/10**
2024	JKBOSE(XII)	Burn Hall School, Sonwar, Srinagar	93.6%
2022	JKBOSE(X)	Burn Hall School, Sonwar, Srinagar	96.6%

Work Experience

Founder's Office Intern - Product and Growth | Cherry Date | *(Dec'25 - Feb'26)*

Objective	• Spearheaded the **zero-to-one market entry** for a safety-centric dating platform to redefine the local ecosystem.
Approach	• Drove **Women First** user acquisition through safety propositions of **Time Compensation** & **Ironclad Safety**.
Impact	• Established an **initial market footprint** by converting offline engagement into a verified early-adopter pipeline.

Research Experience

Depression Classification and Prediction | Prof. Hamim Zafar | **ML - Intern** | Cosmic Lab | *(May'26 - Present)*

Objective	• Build a **multimodal ML pipeline** for Major Depressive Disorder classification & HAMD prediction from MRI.
Approach	• Compared **Linear SVM**, **Random Forest**, **XGBoost**, and a **multimodal GNN** for depression prediction.
Result	• Achieved ROC-AUC 0.68 with multimodal GNN fusion **superseding** the **state-of-the-art** in **MDD** prediction.

Key Projects

Integrated Environmental Strategy | Prof. Abhas Singh | CE-212 (Course Project) | **Grade: A** | *(Aug'25 - Nov'25)*

Problem	• Assess environmental risks across an **810,000 sq ft, 550-bed** medical school, impacting **8,000+ stakeholders**.
Approach	• Developed an integrated **environmental strategy** using **risk prioritisation** and **phased execution** planning.
Results	• Delivered a **47-page implementation roadmap**, projected to significantly **mitigate environmental risks**.

AirSense AI | Project Mentor | Society of Civil Engineers | IIT Kanpur | *(Jun'26 - Present)*

Objective	• Build an air-quality decision system using **PM2.5 forecasting** & real-time **LLM-driven recommendations**.
Approach	• Engineered a **graph-based spatio-temporal** forecasting pipeline for PM2.5 prediction across 31 NCR stations.
Results	• Designed a **multi-agent architecture** with **route-planner** and **notification agents** for real-time support.

Stair-Climbing Robot | Prof. Mohit Subhash Law | TA-212 (Course Project) | Grade: A *(Jan-Apr'26)*

- Engineered a **dual-motor continuous-track stair-climbing robot** in a **6-member team** for payload transport over staircases.
- Designed an **Arduino-controlled bevel-gear lifting mechanism** for adaptive stair negotiation and secure load handling.
- Manufactured & assembled the prototype using **3-D printing**, **drilling**, **milling**, & **soldering**, integrating Mech & Elec subsystems.

Working Model of the London Bridge | Prof. Shashank Shekhar | TA-211 (Course Project) | Grade: A *(Aug-Nov'25)*

- Engineered a **scaled working model** of the London Bridge through **metal fabrication**, replicating its structural mechanism.
- Fabricated **mild-steel** and **galvanized-iron** members using **sheet-metal forming**, **shearing**, and **drilling** for accurate assembly.
- Assembled the structure using **brazing**, **welding**, **sheet metal** and **riveting** to achieve a rigid and durable working final prototype.

Relevant Courses | *(*) Ongoing*

Soil Mechanics (A*)	Environ. & Sustainability (A)	Mechanics of Solids (A)	Introduction to Electronics(A)
Environmental Processes (A)	Fluid Mechanics (B+)	Oscillations and Waves (B+)	Civil Eng. Materials (B+)
Partial Differential Eq. (B+)	(*) AI, ML, DL & Applications	Engineering Graphics	Classical Dynamics

Lab Experience

Soil Mechanics | CE-252 | Prof. Bipin Gupta | Grade: A* *(Jan'26-Apr'26)*

- Characterized geotechnical properties through **Atterberg limits, compaction, permeability** and **shear-strength** tests soils.
- Validated **Terzaghi** consolidation using **oedometer tests** & analyzed shear behavior through direct-shear, triaxial & Mohr's Circle.

Materials for Civil Engineering | CE-243 | Prof. K.V Harish, Prof. Syam Nair | Grade: B+ *(Aug'25-Dec'25)*

- Tested fresh and hardened concrete through **slump, compaction, air-content, penetration** and **compressive-strength** tests.
- Evaluated aggregate properties through **impact, crushing, abrasion, specific gravity, water absorption** and **grading** tests.

Fluid Mechanics | CE-243 | Prof. Guha, Prof. Sahu, Prof. Tripathi | Grade: B+ *(Aug'25-Dec'25)*

- Verified **continuity, momentum conservation** and **Navier-Stokes** assumptions through laboratory experiments and validation.
- Characterized **laminar** & **turbulent flow** using **Reynolds-number estimation, pressure-drop analysis** & flow measurements.

Positions of Responsibility & Extra-Curriculars

- Coordinated **Roadtrip Junoon** as **Organiser**, Events & Competitions, **Antaragni'26**, managing a budget of over **INR 2 Cr+**.
- Secured over **INR 7 Lakh+** in sponsorships, collaborated with **50+ bands**, and conducted **Road Trips** across **7+ metro cities**.
- Elected by a **600+ member electorate** as the **Maintanence Secretary** of Hall III managing a budget of **INR 15 lack** rupees.
- Serving as an **Ex-Officio Member** of the **Anti-Ragging Committee**, contributing to student safety and **grievance handling**.
- Led **11 techincal problem statements** simultaneously as **Hall Captain** during the **Intra-IIT Technological Competition**.